DIRECTIONS *for questions 1 to 5:* Answer these questions on the basis of the information given below.

Vanitha was playing a game in which she drew a 3 × 3 grid and filled it with the digits 1 to 9 as shown in the figure below. She started from any square of the grid and moved either horizontally or vertically (but not diagonally), without passing through any square more than once. While doing this she noted the digits on the squares that she travelled along, from left to right as one number. For example, if she started at the grid with the number 3 and moved to the grid with 8 and finally to the grid with 2, she noted the number 382.

2	8	7
1	3	6
5	9	4

In the number that Vanitha noted, the first digit refers to the digit at the extreme left, while the last digit refers to the digit at the extreme right.

Q1. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

What is the largest number that Vanitha could have noted?

Q2. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

If the last three digits of the number that Vanitha noted were 315, what is the largest number that Vanitha could have noted?

Q3. DIRECTIONS *for question 3:* Select the correct alternative from the given choices.

If Vanitha noted a number with the maximum possible number of digits, and the number formed by the last two digits is X, which of the following cannot be true of X?

a) X is divisible by 19.

b) **X is divisible by 26.**

c) **X is divisible by 31.**

d) **More than one of the above** ✘ **Your answer is incorrect**

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

If in the number that Vanitha noted, the difference in the values of any two successive digits was more than 1, what is the highest possible number that Vanitha could have noted?

Q5. **DIRECTIONS** *for question 5:* Select the correct alternative from the given choices.
If the number that Vanitha noted was not a multiple of 3, then which of the following is/are true of the highest possible number that Vanitha could have noted?

I. It is divisible by 5.

II. The sum of the digits of the number is 41.

III. The number starts with 8.

IV. The sum of the digits of the number is 43.

- a) Only I and IV
- b) **Only II**
- c) **Only III and IV**
- d) **Only IV**

DIRECTIONS *for questions 6 to 10:* Answer these questions on the basis of the information given below.

Three exams – Exam 1, Exam 2 and Exam 3 – were conducted for the students in a school. The three exams had questions of three types – QT1, QT2 and QT3. Each student wrote at least one exam and the questions of each question type appeared in at least one exam. Further, any student who attempted any exam attempted all the questions in the exam.

It is also known that

i. the questions of the type QT2 appeared only in Exam 2, while the number of students who attempted exactly two exams is 27 more than the number of students who attempted only Exam 3.

ii. the number of students who attempted at least one question of type QT1 and the number of students who attempted at least one question of type QT3 are in the ratio 6:7.

iii. the number of students who attempted Exam 1 and the number of students who attempted Exam 3 are in the ratio 4:11.

iv. all the students attempted at least one question of the type QT2.

Q6. **DIRECTIONS** *for question 6:* Type in your answer in the input box provided below the question.
How many students attempted only Exam 2?

Q7. DIRECTIONS *for questions 7 to 10:* Select the correct alternative from the given choices.

Which of the following provides the comprehensive list of exams in which at least one question of type QT3 could have appeared?

- a) Exam 2
- b) **Exam 1, Exam 3**
- c) **Exam 2, Exam 3**
- d) **Exam 1, Exam 2, Exam 3**

Q8. DIRECTIONS *for questions 7 to 10:* Select the correct alternative from the given choices.
What BEST can be said about the number of students who attempted at least one question of type QT1?

- a) It lies between 30 and 35.
- b) **It lies between 35 and 40.**
- c) **It is less than 25.**
- d) **It lies between 40 and 45.**

Q9. DIRECTIONS *for questions 7 to 10:* Select the correct alternative from the given choices.

For each type of question, the total number of questions of that type across the three exams is the same as the number of students who attempted that type of question across the three exams. No question appeared in more than one exam.

If the number of questions in Exam 1, Exam 2 and Exam 3 are in the ratio 2 : 5 : 3, what is the maximum possible number of questions of type QT1 in Exam 1?

a) 36

b) **23**

c) **35**

d) **24**

Q10. **DIRECTIONS** *for questions 7 to 10:* Select the correct alternative from the given choices.
For each type of question, the total number of questions of that type across the three exams is the same as the number of students who attempted that type of question across the three exams. No question appeared in more than one exam.

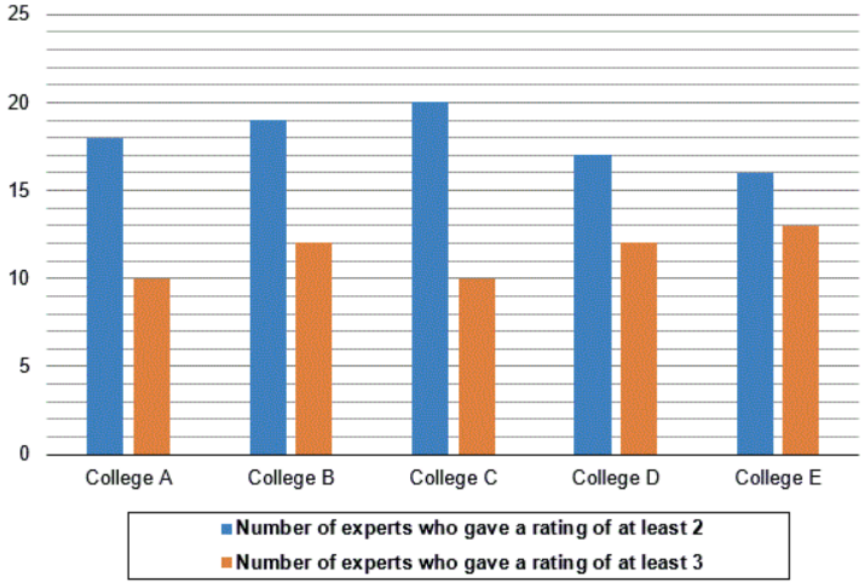
If the number of questions in Exam 1, Exam 2 and Exam 3 are in the ratio 2 : 5 : 3, which of the following options BEST represents the difference between the number of questions of type QT3 in Exam 1 and the number of questions of type QT1 in Exam 3?

- a) Exactly 12
- b) **Either 24 or 36**
- c) **Lies between 1 and 24**
- d) **Exactly 15**

DIRECTIONS *for questions 11 to 15:* Answer these questions on the basis of the information given below.

In a survey to find the best college in a city, twenty-five experts were asked to rate each of five colleges, College A through College E. Each college was given a rating of 1, 2, 3, 4 or 5 by each of the experts and the ratings that each expert gave to the five colleges were not necessarily distinct. It is known that for each college, the number of experts who gave a rating of 1 to that college was less than four-fifths but more than two-fifths the number of experts who gave a rating of 5 to that college.

The graph below provides, for each college, the number of experts who gave a rating of at least 2 and the number of experts who gave a rating of at least 3 for that college.



Q11. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
What is the minimum number of experts who could have given a rating of 5 to College D?

a) 11

b) **9** ✘ Your answer is incorrect

c) **10**

d) **12**

Q12. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
What is the maximum number of experts who could have given a rating of 4 to any of the five colleges?

a) **10** ✓ **Your answer is correct**

b) **9**

c) **12**

d) **11**

Q13. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

Which of the following options BEST represents the colleges for which it is possible that the average of all the ratings obtained by that college is the minimum among the five colleges?

a) **College A, College B, College E**

b) **College A, College B, College C**

c) **College B, College C, College E**

d) **College A, College C**

Q14. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
The number of experts who gave a rating of 5 to the five colleges was distinct. Further, for at least one of the colleges, the sum of the number of experts who gave a rating of 1 and those who gave a rating of 2 was the same as the number of experts who gave a rating of 5.

What BEST can be said about the number of experts who gave a rating of 5 to College E?

- a) **Exactly 12**
- b) **Either 12 or 13**
- c) **Exactly 13**
- d) **Either 11 or 12**

Q15. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

The number of experts who gave a rating of 5 to the five colleges was distinct. Further, for at least one of the colleges, the sum of the number of experts who gave a rating of 1 and those who gave a rating of 2 was the same as the number of experts who gave a rating of 5.

What BEST can be said about the average of all the ratings received by College B?

a) **It is at least 2.88 and at most 3.04.**

b) **It is at least 2.96 and at most 3.04.**

c) **It is at least 2.88 and at most 3.12.**

d) **It is at least 2.96 and at most 3.12.**

Six persons, Ron, Saul, Tim, Usain, Vin and Walt, went to a shop to purchase a certain number of pens. Before they purchased the pens, the six persons had ₹100, ₹200, ₹300, ₹400, ₹500, and ₹600, not necessarily in the same order. Each person purchased a distinct number of pens, and the six persons purchased pens only with the amounts that they had.

After the six persons purchased the pens, they realized that the amounts left with some of them will not be enough to pay for the cab fare back to their respective homes. So, they redistributed the total amount left with them among themselves, such that the amount with each person after the redistribution was exactly enough for the cab fare to their respective homes. The cab fares for the six persons, Usain, Tim, Saul, Ron, Vin and Walt, to their respective homes are ₹43, ₹78, ₹63, ₹159, ₹172 and ₹145, respectively.

It is also known that

i. the cost of each pen was exactly ₹45 and the maximum number of pens that any person purchased was 10.

ii. after purchasing the pens but before redistributing the amounts, Ron had ₹185.

iii. because of the redistribution, the amount with Walt reduced but the amount with Vin increased by more than ₹100 in the redistribution.

iv. before purchasing the pens, Vin had more than ₹100 but had half the amount that Tim had; immediately after the six persons purchased the pens, Vin still had half the amount that Tim had.

v. Saul purchased exactly four pens.

Q16. **DIRECTIONS** *for question 16:* Select the correct alternative from the given choices.
How many pens did Walt purchase?

a) **10**

b) **9**

c) **4**

d) **2**

Six persons, Ron, Saul, Tim, Usain, Vin and Walt, went to a shop to purchase a certain number of pens. Before they purchased the pens, the six persons had ₹100, ₹200, ₹300, ₹400, ₹500, and ₹600, not necessarily in the same order. Each person purchased a distinct number of pens, and the six persons purchased pens only with the amounts that they had.

After the six persons purchased the pens, they realized that the amounts left with some of them will not be enough to pay for the cab fare back to their respective homes. So, they redistributed the total amount left with them among themselves, such that the amount with each person after the redistribution was exactly enough for the cab fare to their respective homes. The cab fares for the six persons, Usain, Tim, Saul, Ron, Vin and Walt, to their respective homes are ₹43, ₹78, ₹63, ₹159, ₹172 and ₹145, respectively.

It is also known that

i. the cost of each pen was exactly ₹45 and the maximum number of pens that any person purchased was 10.

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iv. before purchasing the pens, Vin had more than ₹100 but had half the amount that Tim had; immediately after the six persons purchased the pens, Vin still had half the amount that Tim had.

v. Saul purchased exactly four pens.

Q16. **DIRECTIONS** *for question 16:* Select the correct alternative from the given choices.
How many pens did Walt purchase?

a) **10**

b) **9**

c) **4**

d) **2**

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

What is the amount (in ₹) left with Usain immediately after he purchased the pens?

Q18. **DIRECTIONS** *for question 18:* Select the correct alternative from the given choices.
Which of the following provides the comprehensive list of all persons with whom the amounts with them increased because of the redistribution?

- a) **Ron, Tim, Saul, Vin**
- b) **Usain, Vin**
- c) **Saul, Tim, Vin**
- d) **Saul, Usain, Vin**

Q19. DIRECTIONS *for question 19:* Type in your answer in the input box provided below the question.

What is the total amount (in ₹) left with Saul and Tim combined immediately after the six persons purchased the pens?

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.
What is the total number of pens purchased by Ron and Vin combined?

- a) **12**
- b) **4**
- c) **10**
- d) **5**